Predicting game completion time

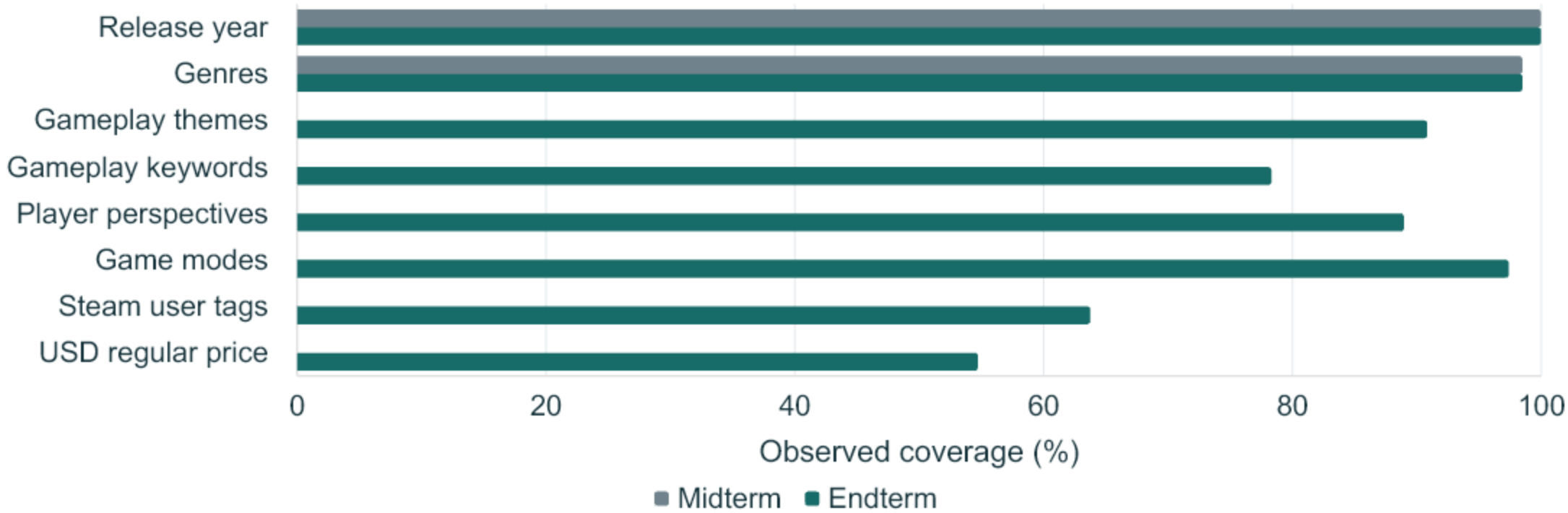
Can game metadata help a player plan a time budget?

Machine Learning Algorithms Endterm, SE2424

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Real feature expansion on the same 5,229 games



2,861 observed USD prices and 3,333 games with Steam tags

The ZIP contains collection code, raw responses, conclusions and defence notes.

Saved labels, split and collection code

4,183 training games

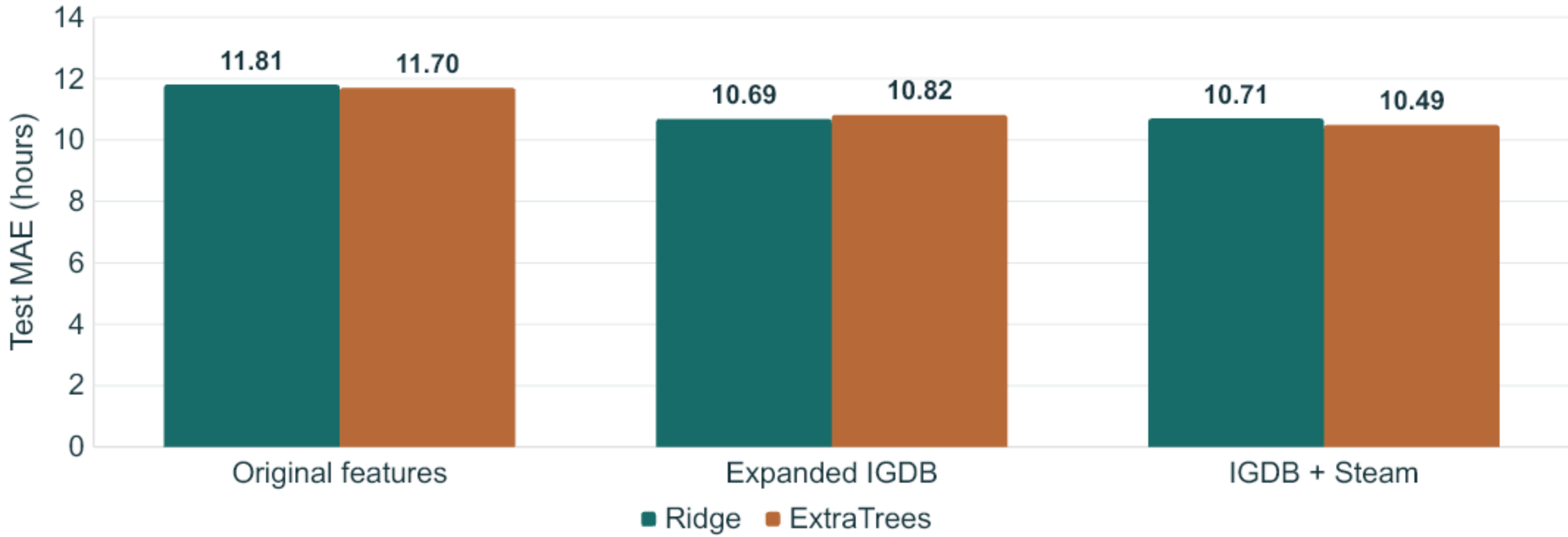
1,046 test games

Step	Evidence in the ZIP
Collect real observations	src/collect.py and reference/midterm_source/src/
Preserve game IDs and target	IGDB main-story hours from the Midterm snapshot
Preserve the saved split	data/clean/fixed_split.csv matches the original bytes
Fit preprocessing inside CV	Imputation, scaling, vocabularies and studio frequency

Three training folds, seed 42. Both ensemble searches use training CV MAE.

The test was already seen at Midterm. It supports comparison between stages.

What the additional feature types contribute



Estimator settings stay fixed within each series. Lower MAE is better.

Steam helps ExtraTrees here. Its additional gain for Ridge is unclear.

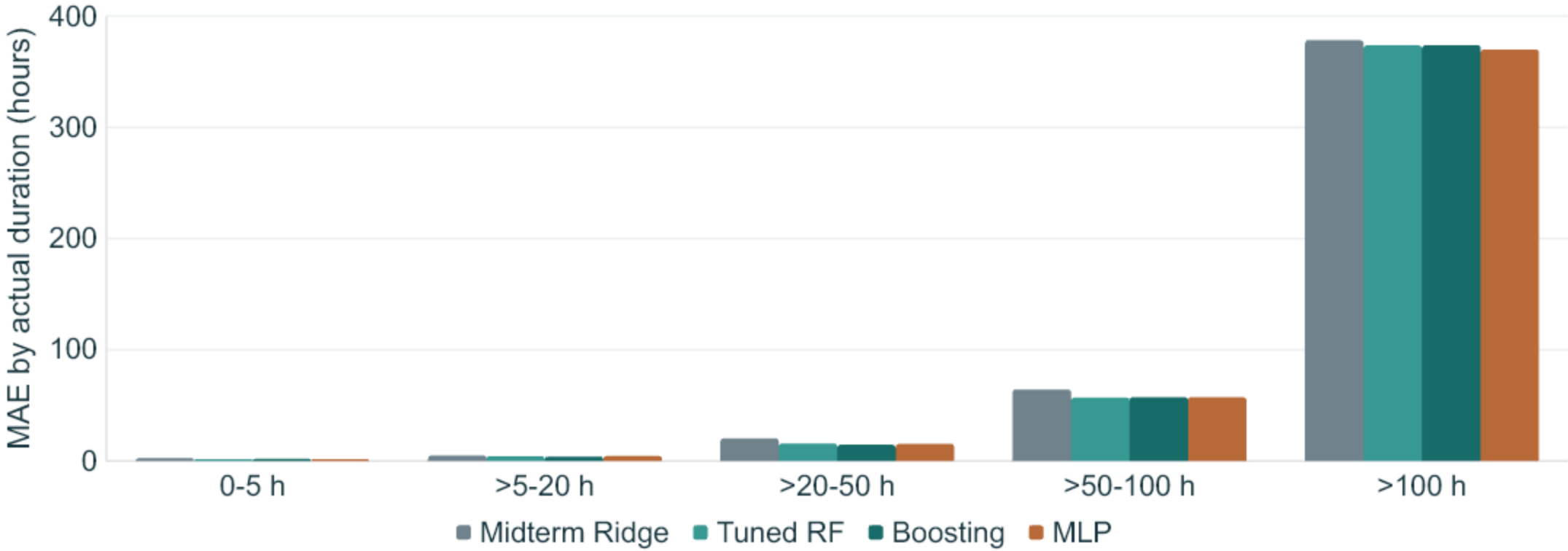
Training-CV selection and tuned ensembles

Model	CV MAE ± SD (h)	Test MAE (h)	Test R^2
Median baseline	11.075 ± 0.716	13.01	-0.041
Ridge original (refit)	10.074 ± 0.680	11.81	-0.012
Ridge expanded	9.157 ± 0.680	10.71	0.024
ExtraTrees expanded	8.772 ± 0.743	10.49	0.022
Tuned random forest	8.636 ± 0.642	10.37	0.022
Tuned boosting	8.639 ± 0.602	10.26	0.023
Tuned MLP	9.016 ± 0.637	10.41	0.036

RF: 8 settings. Boosting: 8 settings. MLP: 4 settings.

Selected by training CV: Random forest (near tie)

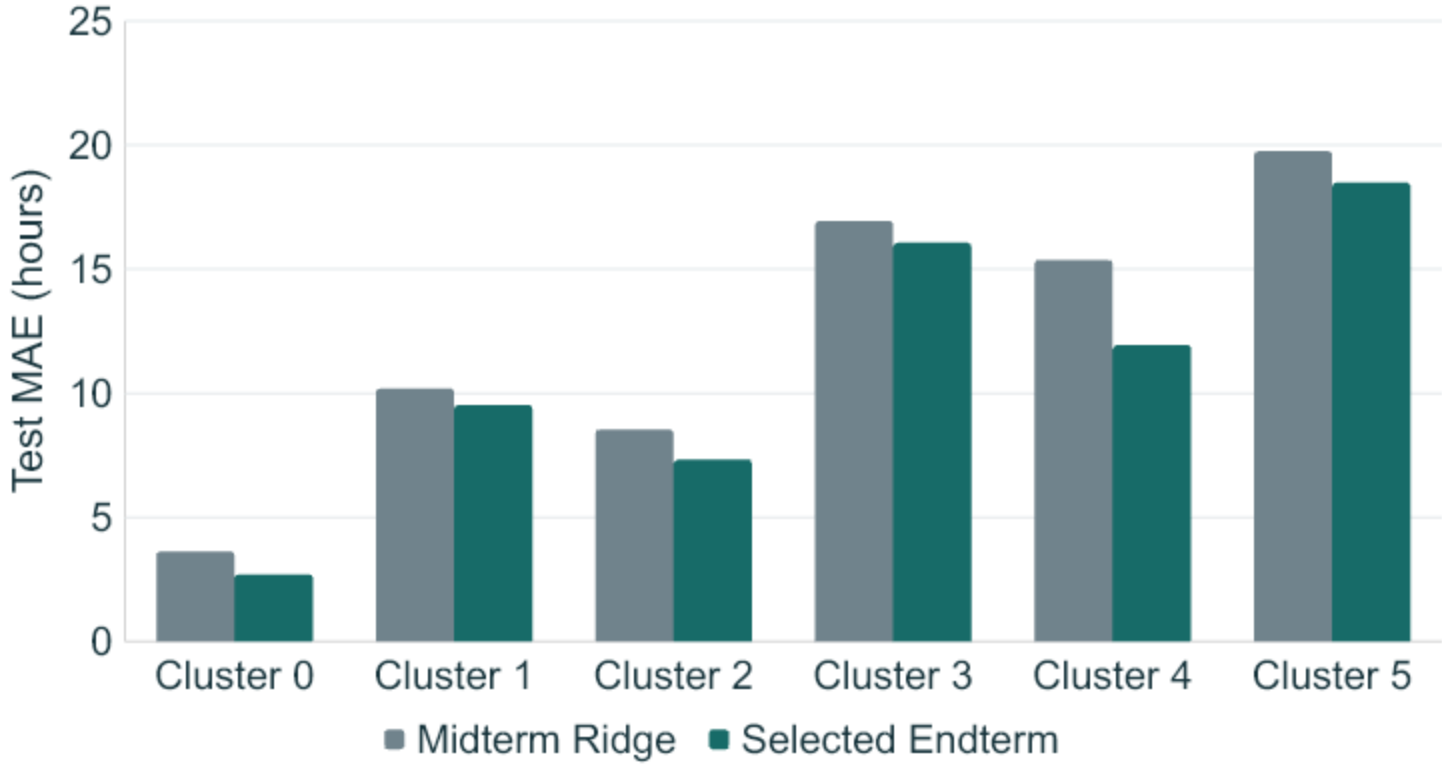
Where different models fail



RF vs boosting: RF has lower MAE at 0-5 h, boosting at >5-20 h.

RF / boosting MAE: 1.80 / 1.91 h (short), 4.30 / 4.14 h (5-20 h).

PCA groups help describe model errors



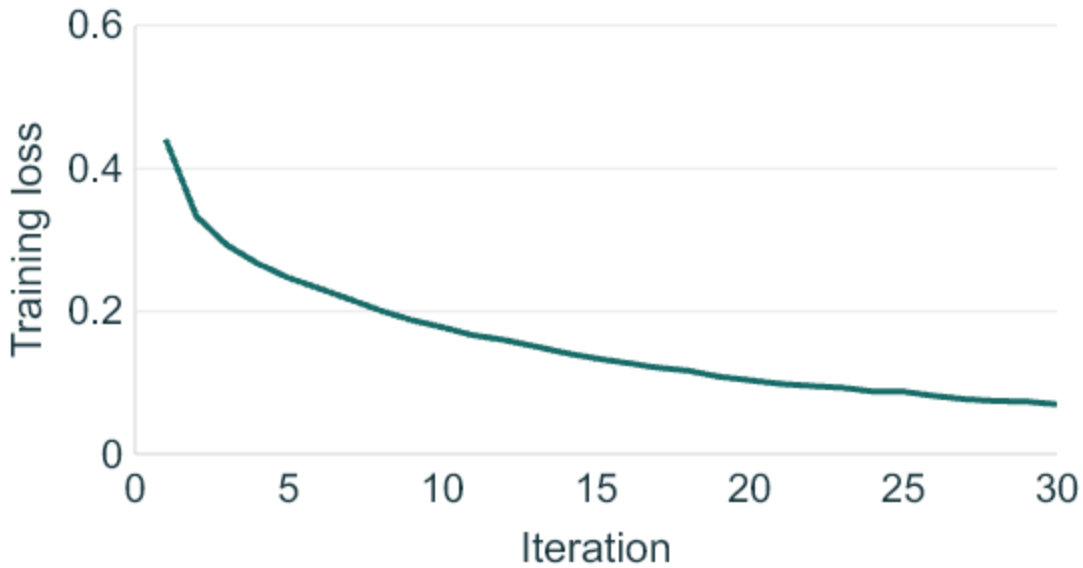
65 PCA components
85.1% variance retained

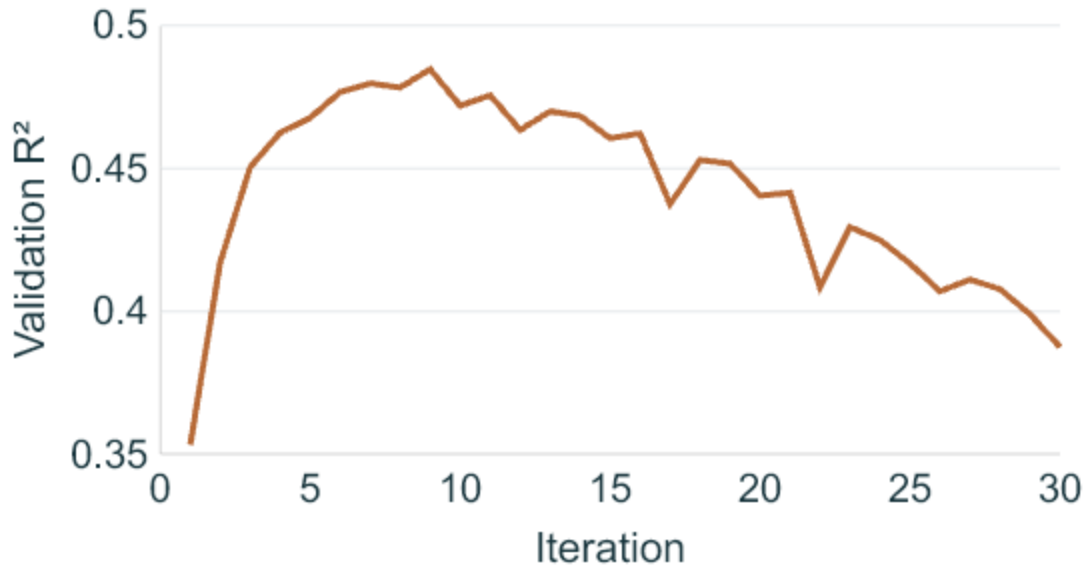
k = 6
Silhouette 0.078
Seed ARI 0.970

The Novelist (cluster 0): Drama, Stealth, Single player.

Actual 2.05 h. Midterm 4.95 h. Endterm 2.87 h. Groups overlap (silhouette 0.078).

MLP training and validation progress





Hidden layers 64 / 32. L2 0.1. Best validation at iteration 9.

Training stops at 30 iterations and restores the best validation weights.

MLP test MAE 10.41 h. Selected model 10.37 h.

Training loss uses standardized log targets. Validation R² uses the same transformed target.

Conclusions and the Final experiment

10.37 h mean absolute test error
20.3% lower than the median baseline

Real metadata improves rough estimates. Long and open-ended labels remain difficult.

Final source: IGDB descriptions linked by the same game ID.
Compare structured inputs, TF-IDF text and their combination using training CV MAE.

Limits: current metadata, selective Steam coverage, uncertain labels and prior test exposure.